

Selective Query-Side Expansion for Long-Horizon Agent Memory Retrieval

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Abstract

Long-horizon software-engineering agents store prior experience as execution traces, tool calls, terminal outputs, and error messages, while later retrieval requests are often phrased as natural-language questions. This representational mismatch can prevent relevant memories from appearing in the retrieved context. We study Selective Query-Side Expansion (SQE), a retrieval-time method that expands only low-confidence queries into hypothetical execution traces and paraphrases, retrieves each variant against the unchanged memory index, and combines ranked lists with Reciprocal Rank Fusion. On a 5,000-episode SWE-smith memory store with 500 query-memory pairs, SQE obtains 69.4% Recall@5, compared with 69.8% for dense retrieval and 64.0% for hybrid RRF. Across 8 independently rebuilt memory-index seeds, SQE averages 69.4% Recall@5, compared with 68.5% for dense retrieval, 69.2% for always-on expansion, and 68.9% for the executed random-gating budget control. The current evidence supports SQE as a cost-aware retrieval variant and shows higher Recall@5 than hybrid RRF, but it does not establish a practically meaningful advantage over dense retrieval or a clear advantage over the executed random-gating budget control. Downstream Pass@1 evaluation and human-audited query labels remain necessary before making end-to-end agent-performance claims.

1 Introduction

Software-engineering agents accumulate long trajectories of observations, commands, file edits, tests, and error traces. These records are useful only if future agents can retrieve the relevant fragment when facing a related problem. The retrieval problem is difficult because the memory and query distributions are not written in the same style: a stored memory may contain a traceback or a patch diff, while the future query may ask how an off-by-one line-number bug was fixed.

We study SQE as a retrieval-time intervention. The memory-writing pipeline remains unchanged. At retrieval time, the method first runs dense retrieval [4] and treats the top-1 score as a confidence signal. If the score is below a threshold, SQE generates execution-style hypothetical traces and paraphrased queries, retrieves

with these variants, and fuses the resulting rankings with Reciprocal Rank Fusion [1]. The method is related to HyDE [2], but is specialized to software-agent traces and is applied selectively to bound generation cost.

Contributions. First, we formulate the trace-query mismatch problem for long-horizon software agent memory. Second, we implement a selective expansion pipeline that combines hypothetical traces, paraphrases, dense retrieval, sparse retrieval, and RRF. Third, we provide a verified retrieval study on SWE-smith trajectories with 8 independently rebuilt memory-index seeds. Fourth, we identify the current limitations: the confidence gate is weakly calibrated, the evaluation uses generated retrieval queries, and downstream Pass@1 has not yet been measured.

2 Related Work

SQE builds on standard dense and sparse retrieval methods. Dense passage retrieval maps queries and documents into a shared embedding space, while BM25 uses lexical term matching and remains a useful sparse baseline [4, 5]. The SQE experiments keep both indexes fixed and modify only the query side, so the method can be evaluated as a retrieval-time intervention rather than a memory-writing change.

Query expansion methods address vocabulary mismatch by rewriting or augmenting the query before retrieval. HyDE generates a hypothetical document and embeds it for zero-shot dense retrieval [2]. SQE uses the same broad idea of hypothetical query-side expansion, but changes the unit of expansion to software-agent execution traces and applies expansion only when the initial retrieval score is low. The generated traces are retrieval probes rather than claims that those executions actually occurred. Reciprocal Rank Fusion provides a simple way to combine ranked lists from the original query, generated traces, paraphrases, and sparse retrieval without training a ranker [1].

For software-engineering agents, retrieval quality is only an intermediate signal. SWE-bench-style evaluation measures whether an agent can produce a patch that resolves a real issue [3]. The present paper therefore reports retrieval metrics as a controlled diagnostic and

treats Pass@1 task success as required future evidence, not as an implied result.

3 Method

Let $M = \{m_i\}_{i=1}^N$ be a memory store of execution traces. Given a query q , dense retrieval returns a ranked list $R_d(q)$ and a top-1 similarity score $s_1(q)$. SQE expands the query if $s_1(q) < \tau$. For expanded queries, an LLM generates K hypothetical traces and P paraphrases. Each variant is retrieved against the same dense and BM25 sparse indexes [5]. The ranked lists are fused using RRF:

$$\text{score}(m) = \sum_j \frac{1}{k + \text{rank}_j(m)}.$$

The current implementation uses $\tau = 0.65$, $K = 2$, $P = 2$, and $k = 60$.

Figure 1 shows the retrieval path. The high-confidence branch uses the initial dense list. The low-confidence branch generates trace probes and paraphrases, retrieves each query variant, and sends all ranked lists to RRF.

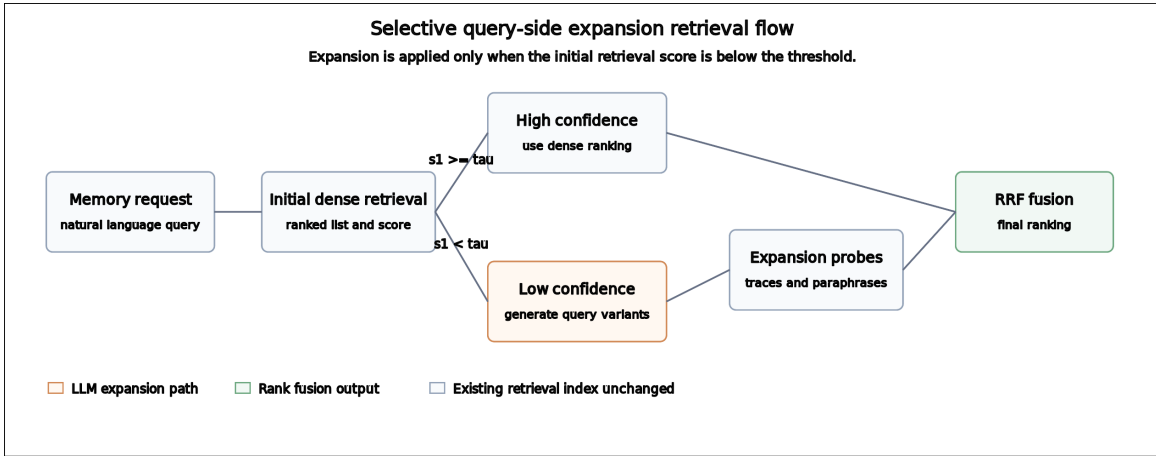


Figure 1: SQE retrieval flow. The method changes only the query side at retrieval time; the memory store and index are left unchanged.

4 Experimental Setup

Table 1 summarizes the current retrieval benchmark. The experiment uses generated natural-language queries paired with target memories. This setup isolates retrieval, but it does not yet establish downstream task success. A complete evaluation requires verified human-audit labels and agent Pass@1 evaluation on SWE-bench-style tasks [3]. Random-Gated-Expansion is an executed control: it runs the same expansion-and-retrieval code as SQE, but replaces the confidence gate with seeded random expansion decisions at a comparable expansion rate and stores the resulting per-query rows in JSONL.

Table 1: Experiment configuration and release placeholders.

Item	Value
Memory episodes	5000
Evaluation queries	500
Dataset split	tool
Evaluation source	memory
Dataset seed	42
Embedding model	BAAI/bge-m3
Generator model	Qwen3.6-35B-A3B via vLLM
Index	FAISS dense + BM25 sparse
Fusion	Reciprocal Rank Fusion
GitHub URL	TODO: add public repository URL
Hugging Face dataset	TODO: add Hugging Face dataset URL

5 Results

We treat the 8-seed aggregate in Table 3 as the primary retrieval evidence. Table 2 is retained as a detailed audit anchor because its per-query rows, cost traces, figures, and case analyses are all regenerated by the artifact pipeline.

In the seed-42 run, SQE changes Recall@5 by -0.4 percentage points relative to dense-only retrieval and +5.4 percentage points relative to hybrid RRF in the current run. Recall@1 is lower than dense retrieval. This indicates that the current SQE configuration has higher Recall@5 than hybrid RRF, but does not improve the strongest dense baseline. The always-on ablation shows that expanding every query can hurt Recall@1, so selectivity is mainly a cost-control mechanism in this run.

5.1 Multi-seed retrieval aggregate

The aggregate uses 8 independently sampled memory stores of 5,000 episodes each and rebuilt indices. SQE has slightly higher mean Recall@5 than dense retrieval, but the paired effect against dense retrieval is small and Recall@1 remains below dense retrieval. The result is therefore evidence for a modest retrieval-only candidate-set difference under this setup, not strong evidence of an end-to-end agent improvement. Figure 2 reports the seed-42 method comparison, while Table 4 reports the paired aggregate uncertainty.

The paired bootstrap over all 4,000 query rows estimates a small Recall@5 improvement over dense retrieval: the Recall@5 difference against dense retrieval is 0.9 points with a 95% interval of [0.1, 1.7]. The same test shows larger gains over hybrid RRF and trace-only expansion, but not a clear advantage over the executed random-gating budget control. At the query level, SQE recovers 140 top-5 targets missed by Dense-Only, but loses 105 targets that Dense-Only retrieved, for a net change of 35 over 4,000 paired queries. Table 5 reports the corresponding win/loss counts.

Table 6 lists deterministic examples from the seed-42 retrieval rows. They are illustrative only and are not additional aggregate metrics.

5.2 Expansion cost

The completed ablations estimate generation cost by counting LLM calls rather than provider-reported tokens. Selective-QE uses fewer expansion calls than always-on expansion because it expands 46.0% of queries. The measured-token reruns below provide provider-reported token usage for all expansion ablations. Table 7 reports estimated calls, and Table 8 reports the measured-token reruns.

5.3 Measured token cost

These reruns execute the expansion methods without the expansion cache after adding usage logging for the OpenAI-compatible API. They are reported as cost measurements; the 500-query retrieval results above remain the main effectiveness comparison.

5.4 Paired uncertainty

The confidence intervals in Table 9 are paired bootstrap intervals over the 500 current queries. They should be treated as diagnostics, not final statistical evidence, because the evaluation set is small and generated from the same trace corpus.

Table 2: Seed-42 retrieval results used as the detailed audit anchor.

Method	Recall@1	Recall@5	Recall@10	MRR	Expansion %
Dense-Only	49.0	69.8	75.4	57.4	–
BM25-Only	17.6	31.4	35.8	23.3	–
Hybrid-RRF	35.4	64.0	73.2	47.4	–
Paraphrases-Only	47.0	68.2	75.0	56.0	100.0
HyDE-Traces-Only	36.2	64.0	70.4	47.9	100.0
Always-Expand	42.2	68.0	75.6	53.2	100.0
Random-Gated-Expansion	46.0	69.0	75.8	55.3	47.0
Selective-QE	47.2	69.4	75.4	56.4	46.0

Table 3: Aggregate retrieval results over independent memory-index seeds.

Method	Seeds	Recall@1	Recall@5	Recall@10
Dense-Only	8	45.5 $\pm$ 3.1	68.5 $\pm$ 2.7	74.2 $\pm$ 2.5
Hybrid-RRF	8	32.5 $\pm$ 1.5	63.5 $\pm$ 2.5	71.8 $\pm$ 2.6
Always-Expand	8	39.8 $\pm$ 3.1	69.2 $\pm$ 3.4	75.9 $\pm$ 2.3
Random-Gated-Expansion	8	42.6 $\pm$ 2.9	68.9 $\pm$ 3.0	75.0 $\pm$ 2.6
Selective-QE	8	44.5 $\pm$ 3.4	69.4 $\pm$ 2.7	75.3 $\pm$ 2.2

Table 4: Paired bootstrap tests over all independent memory-index seeds.

Baseline	Queries	$\Delta R@5$	95% CI	Bootstrap p
Dense-Only	4000	+0.9	[+0.1, +1.7]	0.015
Hybrid-RRF	4000	+5.9	[+4.8, +7.0]	0.000
Always-Expand	4000	+0.1	[-0.4, +0.7]	0.315
Random-Gated-Expansion	4000	+0.5	[-0.2, +1.1]	0.085
Paraphrases-Only	4000	+1.1	[+0.3, +1.9]	0.005
HyDE-Traces-Only	4000	+5.6	[+4.7, +6.6]	0.000

Table 5: Query-level top-5 wins and losses for SQE against Dense-Only.

Split	N	Both hit	SQE win	SQE loss	Both miss	Net
seed 42	500	334	13	15	138	-2
seed 43	500	314	18	18	150	0
seed 44	500	347	17	13	123	4
seed 45	500	335	18	10	137	8
seed 46	500	350	16	11	123	5
seed 47	500	320	17	15	148	2
seed 48	500	322	22	11	145	11
seed 49	500	313	19	12	156	7
All	4000	2635	140	105	1120	35

Table 6: Seed-42 examples where SQE and Dense-Only differ at top-5.

Case	Query ID	Dense rank	SQE rank	Expanded	Query
SQE gain	q_0014	>10	3	yes	What regular expression pattern was used to define the digit group variable?
SQE gain	q_0063	8	3	yes	Did the test script confirm that the parser correctly handles NULL comparisons...
SQE loss	q_0002	1	7	yes	Did the test script confirm that the fix for the FOO value was successful?
SQE loss	q_0031	1	7	yes	What was the output when I ran the simple test script to verify the initial cal...

Table 7: Estimated expansion cost for each retrieval method.

Method	Expansion %	Estimated LLM calls/query
Dense-Only	–	0.00
BM25-Only	–	0.00
Hybrid-RRF	–	0.00
Paraphrases-Only	100.0	1.00
HyDE-Traces-Only	100.0	2.00
Always-Expand	100.0	3.00
Random-Gated-Expansion	47.0	1.41
Selective-QE	46.0	1.38

Table 8: Provider-reported token usage for measured-token reruns.

Method	Expanded	Calls/q	Prompt tok/q	Total tok/q	Latency s/q
Selective-QE	46.0	1.38	209.2	405.1	2.77
Always-Expand	100.0	2.99	455.5	899.3	5.37
HyDE-Traces-Only	100.0	2.00	367.0	758.2	3.99
Paraphrases-Only	100.0	1.00	88.5	131.6	1.02
Random-Gated-Expansion	47.0	1.41	214.6	423.9	2.40

Table 9: Seed-42 paired bootstrap uncertainty.

Comparison	Δ Recall@5	95% CI	Bootstrap p
Selective-QE – Dense-Only	-0.4	[-2.6, 1.8]	0.777
Selective-QE – Hybrid-RRF	5.4	[2.2, 8.6]	0.002
Selective-QE – Always-Expand	1.4	[-0.4, 3.2]	0.151
Selective-QE – Random-Gated-Expansion	0.4	[-1.6, 2.4]	0.738
Selective-QE – Paraphrases-Only	1.2	[-1.0, 3.4]	0.314
Selective-QE – HyDE-Traces-Only	5.4	[2.6, 8.2]	0.000

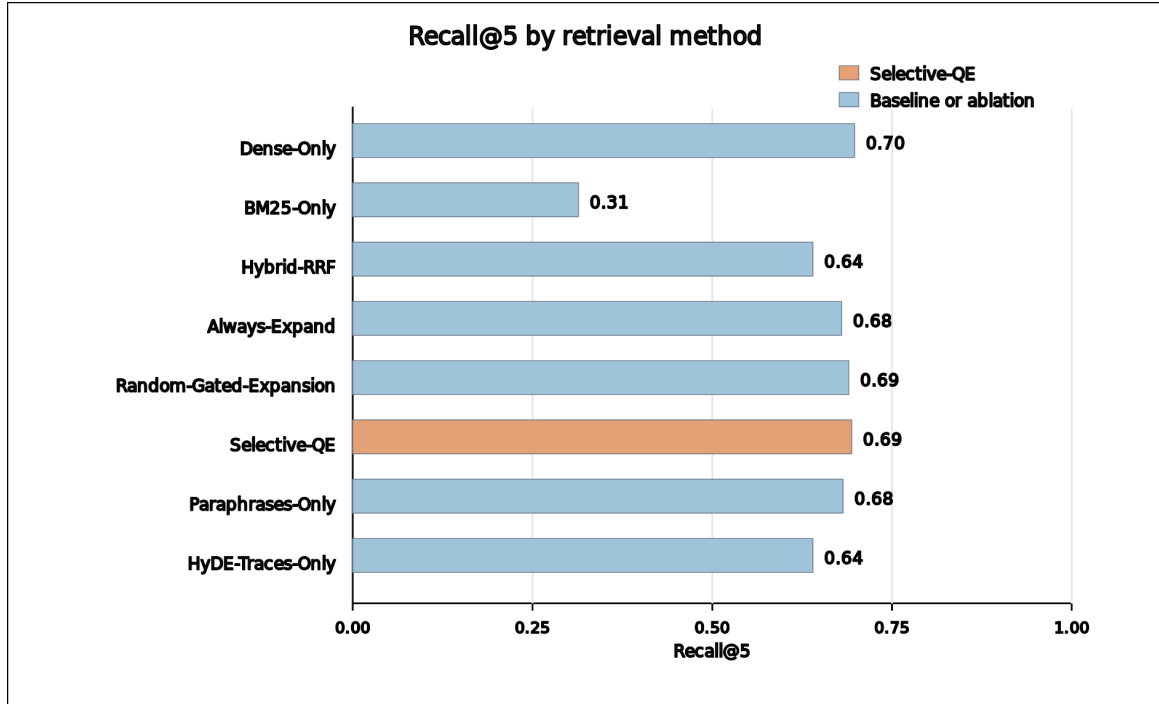


Figure 2: Recall@5 comparison across completed retrieval methods for seed 42. SQE is highlighted; the other bars are baselines or ablations.

6 Gate Diagnostics

Table 10 shows that the gate expands 230 of 500 queries. In the current run, expanded queries do not outperform non-expanded queries at Recall@5, so top-1 dense score is not yet a well-calibrated selector. This is a central weakness for the current method claim. Stronger gates should include top-1/top-2 margin, dense-sparse agreement, score concentration, or a learned policy.

6.1 Post-hoc threshold sweep

Table 11 and Figure 4 reuse the completed dense-only and always-expand result files, so the sweep is diagnostic rather than a tuned validation result. It indicates that the fixed 0.65 threshold is not optimal on this split, and threshold selection needs a separate validation protocol.

6.2 Split validation check

Table 12 reports a minimal guard against all-query tuning: we select a threshold on the first half of the query IDs and report it on the second half. This split is small and should not be treated as final evidence, but it makes the calibration weakness explicit.

6.3 Multi-seed gate validation

Table 13 repeats the split-threshold procedure for each completed independent memory-index seed. It recombines only executed dense-only and always-expand rows. The selected thresholds change mean test Recall@5 from 68.2% for dense retrieval to 69.2%, while expanding 45.6% of held-out queries, but the result is still too small to support a strong gate-calibration claim. A paired held-out test over 2000 query/seed pairs estimates a 1.1 point Recall@5 difference with a 95% interval of [0.1, 2.1] and sign-flip $p = 0.037$. Table 14 reports the paired test.

6.4 Cross-seed top-1 gate

Table 15 selects a top-1-score threshold on two independent memory-index seeds and evaluates it on the held-out seed. It recombines only executed dense-only and always-expand rows. Leave-one-seed-out top-1 threshold selection obtains 69.1% Recall@5 versus 68.5% for dense retrieval while expanding 36.9% of queries. The paired difference is 0.6 points with a 95% interval of [-0.1, 1.3]. The interval still crosses zero, so this result is useful as a calibration diagnostic rather than a strong effectiveness claim.

6.5 BM25-agreement gate variants

Table 16 selects among score-only and BM25-agreement gate variants on the first half of each seed and evaluates on the second half. It uses only executed dense-only,

BM25-only, and always-expand rows. The result is diagnostic: the selected variants do not provide enough evidence to remove the gate calibration limitation. Figure 3 shows why the top-1 score is a weak selector in the current experiment.

6.6 Gate headroom

Table 17 recombines executed dense-only and always-expand rows on the second half of each independent memory seed. The oracle column is not an implementable method; it shows the maximum possible Recall@5 if a perfect gate could expand exactly the queries helped by always-on expansion. The small gap between helpful and harmful expansion cases explains why the current confidence gate is difficult to calibrate.

Table 10: Seed-42 gate diagnostic split by expansion decision.

Subset	Count	Hits@5	Misses@5	Recall@5
Expanded	230	142	88	61.7
Not expanded	270	205	65	75.9

Table 11: Seed-42 post-hoc threshold sweep.

Threshold	Expanded	Recall@5	Recall@10
0.45	0/500	69.8	75.4
0.50	1/500	70.0	75.6
0.55	17/500	70.0	75.6
0.60	76/500	70.0	75.8
0.62	130/500	70.2	76.2
0.64	191/500	69.6	75.4
0.65	230/500	69.4	75.4
0.66	275/500	69.8	75.8
0.68	340/500	69.0	75.8
0.70	393/500	69.2	75.8
0.75	478/500	68.0	75.6
0.80	500/500	68.0	75.6

Table 12: Seed-42 split validation check for threshold selection.

Policy	Threshold	Expanded	Recall@5	Recall@10
Dense-only proxy	0.45	0/250	70.4	75.2
Fixed gate	0.65	115/250	71.2	77.6
Validation-selected gate	0.50	0/250	70.4	75.2

Table 13: Held-out gate validation over independent memory-index seeds.

Seed	Threshold	Train R@5	Test R@5	Dense Test R@5	Expansion
42	0.50	69.6	70.4	70.4	0.0
43	0.60	70.0	64.0	63.6	17.2
44	0.75	76.8	72.0	69.2	96.4
45	0.75	71.2	73.6	69.6	97.6
46	0.65	74.4	72.0	71.2	51.6
47	0.45	65.6	68.4	68.4	0.0
48	0.70	68.0	69.2	69.2	82.8
49	0.60	68.4	64.4	63.6	19.6
Mean	–	70.5	69.2	68.2	45.6

Table 14: Paired uncertainty for held-out gate validation.

Scope	Queries	Gate R@5	Dense R@5	Delta	95% CI
42	250	70.4	70.4	+0.0	[+0.0, +0.0]
43	250	64.0	63.6	+0.4	[-0.8, +2.0]
44	250	72.0	69.2	+2.8	[-1.2, +7.2]
45	250	73.6	69.6	+4.0	[+0.4, +7.6]
46	250	72.0	71.2	+0.8	[-2.0, +3.6]
47	250	68.4	68.4	+0.0	[+0.0, +0.0]
48	250	69.2	69.2	+0.0	[-4.0, +4.0]
49	250	64.4	63.6	+0.8	[-2.0, +3.6]
All	2000	69.2	68.2	+1.1	[+0.1, +2.1]

Table 15: Leave-one-seed-out top-1 gate validation.

Test seed	Train seeds	Threshold	Gate R@5	Dense R@5	Delta	Expansion
42	43,44,45,46,47,48,49	0.64	69.6	69.8	-0.2	38.2
43	42,44,45,46,47,48,49	0.62	66.6	66.4	+0.2	27.6
44	42,43,45,46,47,48,49	0.62	72.6	72.0	+0.6	28.8
45	42,43,44,46,47,48,49	0.62	69.8	69.0	+0.8	28.8
46	42,43,44,45,47,48,49	0.62	72.0	72.2	-0.2	29.0
47	42,43,44,45,46,48,49	0.64	67.4	67.0	+0.4	42.6
48	42,43,44,45,46,47,49	0.66	68.6	66.6	+2.0	58.8
49	42,43,44,45,46,47,48	0.64	66.4	65.0	+1.4	41.2
All	–	–	69.1	68.5	+0.6	36.9

Table 16: Held-out BM25-agreement gate variants.

Seed	Policy	Threshold	Test R@5	Dense Test R@5	Expansion
42	score_and_no_bm25_agreement	0.50	70.4	70.4	0.0
43	score_or_no_bm25_agreement	0.65	64.0	63.6	79.6
44	score_only	0.75	72.0	69.2	96.4
45	score_or_no_bm25_agreement	0.70	73.6	69.6	90.4
46	score_only	0.65	72.0	71.2	51.6
47	score_and_no_bm25_agreement	0.45	68.4	68.4	0.0
48	score_and_no_bm25_agreement	0.70	68.8	69.2	55.6
49	score_only	0.60	64.4	63.6	19.6
Mean	–	–	69.2	68.2	49.1

Table 17: Held-out oracle headroom for a perfect expansion gate.

Seed	Dense R@5	Always R@5	Oracle R@5	Helpful	Harmful
42	70.4	70.0	74.8	4.4	4.8
43	63.6	63.6	68.0	4.4	4.4
44	69.2	72.0	76.4	7.2	4.4
45	69.6	73.6	76.0	6.4	2.4
46	71.2	72.4	75.6	4.4	3.2
47	68.4	68.0	73.6	5.2	5.6
48	69.2	68.8	74.0	4.8	5.2
49	63.6	66.0	69.2	5.6	3.2
Mean	68.2	69.3	73.5	5.3	4.2

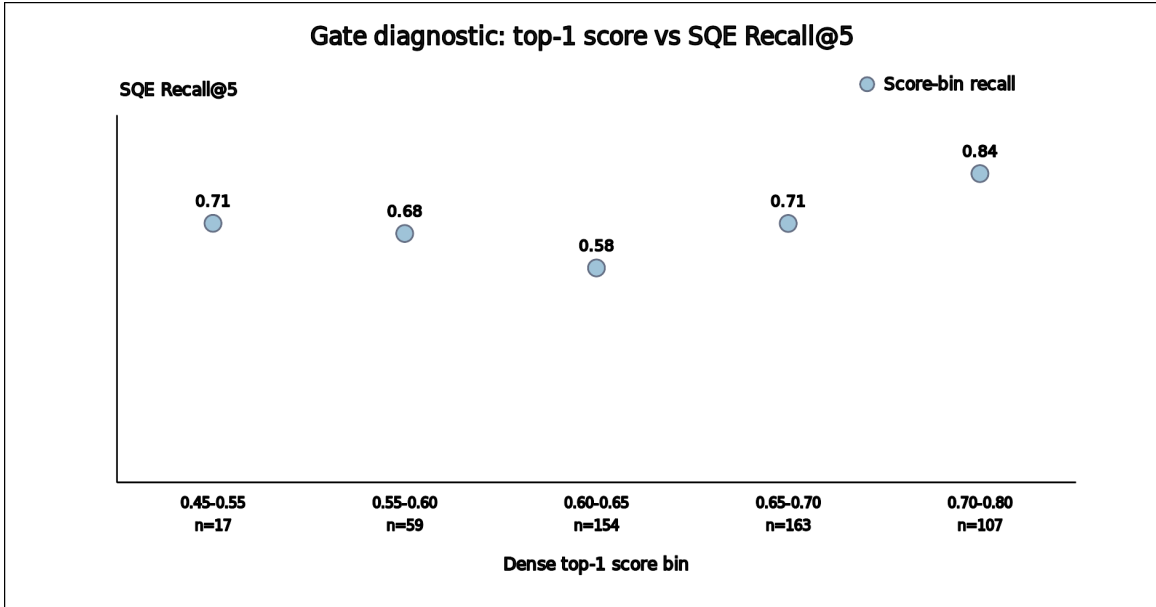


Figure 3: Diagnostic relationship between dense top-1 score bins and final SQE Recall@5. The bins do not separate easy and hard cases cleanly enough for a strong gate claim.

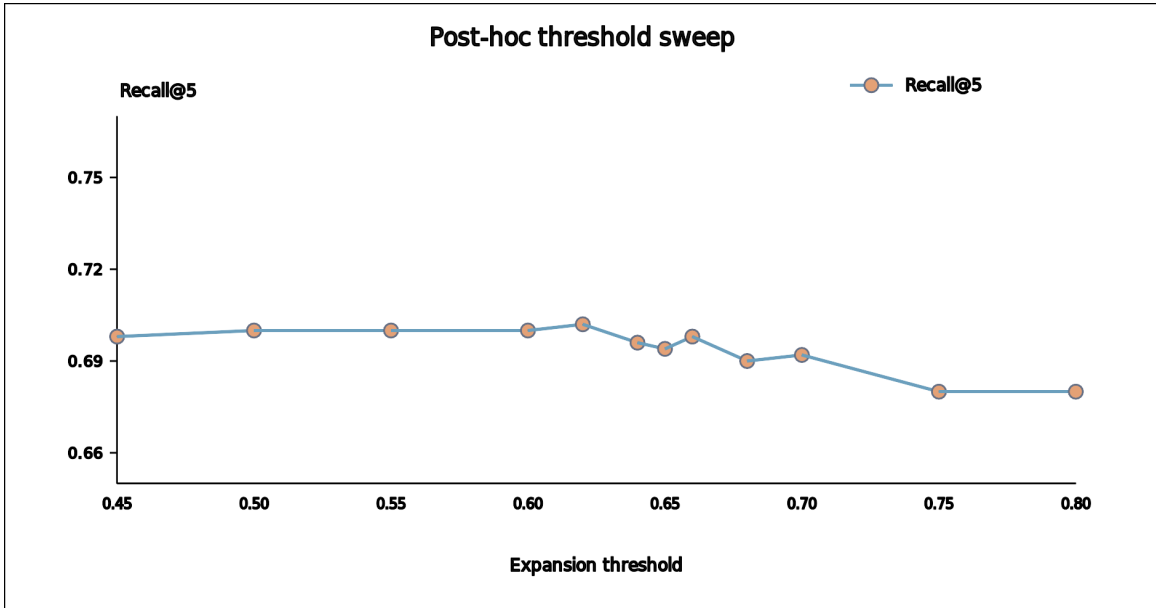


Figure 4: Post-hoc Recall@5 under different expansion thresholds for seed 42. The figure is diagnostic because the threshold is evaluated on completed result rows.

7 Limitations and Next Experiments

The current evidence is preliminary. The retrieval experiment now includes 8 independently rebuilt memory-index seeds, but the improvement over dense retrieval is small and there is no downstream Pass@1 measurement. Simple held-out gate diagnostics using top-1 score, dense margin, score concentration, and BM25 agreement have not removed the calibration weakness. A stronger submission needs externally validated gate calibration, human-audited query labels, and end-to-end agent success measurements.

8 Threats to Validity

Construct validity. Recall@K measures whether the target memory appears in the retrieved set, not whether an agent can use that memory to fix a software task. The current retrieval metrics are therefore intermediate evidence. Pass@1 task success and human query-quality labels are required before claiming agent-level gains.

Data validity. The evaluation queries are generated from the same memory corpus used to define target memories. This gives a controlled retrieval benchmark, but it may not match how users or agents phrase future memory requests. The human-audit packet is prepared, but no reviewer labels have been collected yet.

Method validity. The selective gate uses top-1 dense score, which is weakly calibrated in the current experiments. Several held-out diagnostics test alternative

gates, but their intervals still cross zero. The current results should be read as a cost-controlled retrieval study rather than evidence that the gate is solved.

External validity. The experiments use SWE-smith software-engineering traces, BAAI/bge-m3 embeddings, and Qwen3.6-35B-A3B for expansion. Results may change with other agent logs, embedding models, generators, memory-writing policies, or task domains.

9 Reproducibility

The intended public repository URL is: TODO: add GitHub URL. The intended Hugging Face dataset URL is: TODO: add Hugging Face dataset URL. All scripts used for the current retrieval experiment are in the project directory. Raw seed-42 results are stored under results_500_memory_seed42/, and generated paper artifacts are stored under paper/. Independent memory-index seed results are stored under the results_500_memory_seed42/ through results_500_memory_seed49/ directories. The verifier scripts/07_verify_experiment.py checks that evaluation targets are present in both the memory store and the index, recomputes retrieval metrics from detailed JSONL files, and verifies required paper artifacts. The dataset-release helper scripts/33_prepare_hf_dataset_release.py packages existing memory stores, evaluation pairs, retrieval summaries, diagnostics, and the unlabeled human-audit packet into a local Hugging Face-ready directory. It does not create human labels, downstream Pass@1 rows, or synthetic replacement data. The code-release helper

scripts/34_prepare_github_code_release.py prepares a local GitHub-ready code package with scripts, paper source, configs, documentation, and optional lightweight summaries. It does not upload to GitHub and does not package raw memory stores, detailed query rows, downstream task-outcome rows, or human labels. The table inventory paper/table_inventory.json records the source files for each active table; deprecated diagnostic tables are excluded from the paper claims.

10 Conclusion

SQE is a retrieval-time query-expansion method for long-horizon software-agent memory. In the current experiments, it reduces expansion calls relative to always-on expansion and improves Recall@5 over Hybrid-RRF and trace-only expansion, but it remains essentially tied with dense retrieval and the executed random-gating budget control. The present evidence is best read as a verified retrieval study and a negative calibration result for the current top-1-score gate. A stronger paper requires human-audited query labels and downstream Pass@1 task-success measurements before making agent-level performance claims.

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